

Pandas Full Notes (Beginner to Advanced) - Full Code Examples

1. Introduction & Setup

```
import pandas as pd
print(pd.__version__)
```

2. Series Example

```
import pandas as pd

data = [10, 20, 30, 40]
s = pd.Series(data, index=['a', 'b', 'c', 'd'])

print(s)
print(s['a'])
print(s.mean())
```

3. DataFrame Example

```
import pandas as pd

data = {
    "Name": ["Faruk", "Rahim", "Karim"],
    "Age": [21, 22, 23],
    "Salary": [20000, 25000, 30000]
}

df = pd.DataFrame(data)
print(df)
```

4. Reading & Writing CSV

```
import pandas as pd

df = pd.read_csv("data.csv")
print(df.head())

df.to_csv("output.csv", index=False)
```

5. Data Inspection

```
print(df.head())
print(df.tail())
print(df.info())
print(df.describe())
```

```
print(df.shape)
```

6. Selecting & Filtering

```
print(df["Name"])
print(df[["Name", "Salary"]])

print(df.loc[0])
print(df.iloc[1])

filtered = df[df["Age"] > 21]
print(filtered)
```

7. Handling Missing Data

```
df = df.dropna()

df["Salary"] = df["Salary"].fillna(df["Salary"].mean())

print(df.isnull().sum())
```

8. Sorting & GroupBy

```
sorted_df = df.sort_values("Salary", ascending=False)
print(sorted_df)

grouped = df.groupby("Age")["Salary"].mean()
print(grouped)
```

9. Merge Example

```
df1 = pd.DataFrame({"id": [1, 2], "Name": ["A", "B"]})
df2 = pd.DataFrame({"id": [1, 2], "Salary": [20000, 30000]})

merged = pd.merge(df1, df2, on="id", how="inner")
print(merged)
```

10. Apply & Lambda

```
df["Bonus"] = df["Salary"].apply(lambda x: x * 0.1)
print(df)
```

11. Pivot Table

```
pivot = pd.pivot_table(df, values="Salary", index="Age", aggfunc="mean")
print(pivot)
```

12. Working with Dates

```
df["Date"] = pd.to_datetime(["2024-01-01", "2024-02-01", "2024-03-01"])
df["Year"] = df["Date"].dt.year
df["Month"] = df["Date"].dt.month
print(df)
```

13. Advanced (MultiIndex & Category)

```
df.set_index(["Age", "Name"], inplace=True)
print(df)

df.reset_index(inplace=True)

df["Age"] = df["Age"].astype("category")
print(df.dtypes)
```